Traffic Flow Analysis Using Power BI – Case Study Report

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1. Introduction

This case study focuses on analyzing traffic flow patterns using Power BI. The objective is to understand traffic behaviour across different junctions, identify congestion trends, and predict future traffic volume using time-series analysis.

Efficient traffic management is essential for urban planning, and this project aims to provide data-driven insights to support decision-making.

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2. Problem Statement

Urban traffic congestion leads to delays, increased fuel consumption, and environmental impact. The goal of this project is to:

- Identify peak traffic hours
- Analyze traffic distribution across junctions
- Detect patterns and trends in traffic flow
- Predict future traffic volume
- Provide actionable solutions for traffic management

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3. Dataset Overview

The dataset contains hourly traffic data collected from multiple junctions.

Key Variables:

- **DateTime** – Timestamp of traffic observation
- **Junction** – Location identifier
- **Vehicles** – Number of vehicles recorded

The dataset spans from **2015 to 2017**, providing sufficient data for trend and forecasting analysis.

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4. Data Cleaning & Preparation

To ensure accurate analysis, the following steps were performed:

- Converted DateTime into proper date format
- Created additional columns:
  - Hour
  - Day
  - Month
- Removed missing or null values
- Ensured consistent data types
- Eliminated unnecessary columns

👉 Result: A clean and structured dataset ready for analysis

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5. Data Modeling

A **star schema** was implemented to improve performance and organization.

Fact Table:

- Traffic Dataset (Vehicles, DateTime, Junction)

Dimension Tables:

- Dim_Time (Date, Month, Year)
- Dim_Junction (Junction)

Relationships:

- Traffic[DateTime] → Dim_Time[Date]
- Traffic[Junction] → Dim_Junction[Junction]

Hierarchies:

- Time hierarchy: **Year → Month → Day**

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6. Exploratory Data Analysis (EDA)

Initial analysis revealed important traffic patterns:

- Traffic is lowest during early morning hours
- Peak traffic occurs during daytime
- Junction 1 experiences the highest traffic volume
- Weekdays show higher traffic compared to weekends

These insights guided further analysis and visualization.

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7. Dashboard Development

An interactive dashboard was created using Power BI.

Key Visualizations:

- Line chart (traffic over time)
- Bar chart (traffic by junction)
- Heatmap (hour vs junction)
- KPI cards:
    - Total Vehicles
    - Peak Traffic
    - Average Vehicles

The dashboard allows users to quickly understand traffic behaviour.

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8. Time-Series Analysis

A time-series analysis was performed using a continuous date-based line chart.

Observations:

- Traffic shows a **steady upward trend over time**
- Strong **daily seasonality** is observed
- Traffic patterns are consistent and predictable

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9. Moving Average (DAX)

A 7-day moving average was calculated using DAX to smooth short-term fluctuations.

Benefits:

- Reduces noise in data
- Highlights long-term trends
- Improves clarity for forecasting

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10. Forecasting & Predictive Model

Forecasting was implemented using Power BI's built-in analytics feature.

Features:

- Future traffic prediction
- Confidence interval (95%)
- Trend-based forecasting

Predictive Model:

The model is based on:

- Historical traffic data
- Time-series patterns
- Moving averages

👉 This helps estimate future traffic behaviour.

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11. Model Validation

The model was validated by comparing predicted values with historical data.

Findings:

- Forecast aligns closely with observed trends
- Moving average follows actual data patterns
- Predictions are reliable for trend estimation

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12. Interactivity Features

The dashboard includes advanced interactive features:

- **Slicers:** Date, Hour, Junction
- **Drill-down:** Year → Month → Day
- **Tooltips:** Additional insights on hover

👉 Enables users to explore data in detail

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13. Performance Optimization

To ensure efficiency:

- Used Power BI Performance Analyzer
- Optimized DAX measures
- Removed unused columns

👉 Result: Faster and optimized dashboard performance

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14. Key Insights

- Traffic volume is increasing over time
- Peak traffic occurs during daytime hours
- Junction 1 has the highest congestion
- Traffic patterns are consistent and predictable
- Forecast indicates continued growth in traffic

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15. Proposed Solutions

Based on the analysis, the following solutions are recommended:

- Optimize traffic signal timings during peak hours
- Implement traffic control measures at high congestion junctions
- Use predictive insights for future urban planning
- Improve road infrastructure in high-traffic areas

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16. Limitations

- Dataset does not include weather or event data
- Limited variables restrict deeper analysis
- No real-time traffic data

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17. Future Improvements

Future enhancements may include:

- Integration of weather data
- Inclusion of public event data
- Real-time traffic monitoring
- Advanced machine learning models

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18. Conclusion

This project demonstrates how Power BI can be used for end-to-end data analysis, including data cleaning, modelling, visualization, and forecasting.

The insights derived from this analysis can support better traffic management and planning decisions, ultimately improving urban mobility.